

11.7 Fourier Integral

Fourier series are powerful tools for problems involving functions that are periodic or are of interest on a finite interval only. Sections 11.2 and 11.3 first illustrated this, and various further applications follow in Chap. 12. Since, of course, many problems involve functions that are *nonperiodic and are of interest on the whole x -axis*, we ask what can be done to extend the method of Fourier series to such functions. This idea will lead to “Fourier integrals.”

In Example 1 we start from a special function f_L of period $2L$ and see what happens to its Fourier series if we let $L \rightarrow \infty$. Then we do the same for an *arbitrary* function f_L of period $2L$. This will motivate and suggest the main result of this section, which is an integral representation given in Theorem 1 below.

EXAMPLE 1

Rectangular Wave

Consider the periodic rectangular wave $f_L(x)$ of period $2L > 2$ given by

$$f_L(x) = \begin{cases} 0 & \text{if } -L < x < -1 \\ 1 & \text{if } -1 < x < 1 \\ 0 & \text{if } 1 < x < L \end{cases}$$

The left part of Fig. 280 shows this function for $2L = 4, 8, 16$ as well as the nonperiodic function $f(x)$, which we obtain from f_L if we let $L \rightarrow \infty$,

$$f(x) = \lim_{L \rightarrow \infty} f_L(x) = \begin{cases} 1 & \text{if } -1 < x < 1 \\ 0 & \text{otherwise.} \end{cases}$$

We now explore what happens to the Fourier coefficients of f_L as L increases. Since f_L is even, $b_n = 0$ for all n . For a_n the Euler formulas (6), Sec. 11.2, give

$$a_0 = \frac{1}{2L} \int_{-1}^1 dx = \frac{1}{L}, \quad a_n = \frac{1}{L} \int_{-1}^1 \cos \frac{n\pi x}{L} dx = \frac{2}{L} \int_0^1 \cos \frac{n\pi x}{L} dx = \frac{2}{L} \frac{\sin(n\pi/L)}{n\pi/L}.$$

This sequence of Fourier coefficients is called the **amplitude spectrum** of f_L because $|a_n|$ is the maximum amplitude of the wave $a_n \cos(n\pi x/L)$. Figure 280 shows this spectrum for the periods $2L = 4, 8, 16$. We see that for increasing L these amplitudes become more and more dense on the positive w_n -axis, where $w_n = n\pi/L$. Indeed, for $2L = 4, 8, 16$ we have 1, 3, 7 amplitudes per “half-wave” of the function $(2 \sin w_n)/(Lw_n)$ (dashed in the figure). Hence for $2L = 2^k$ we have $2^{k-1} - 1$ amplitudes per half-wave, so that these amplitudes will eventually be everywhere dense on the positive w_n -axis (and will decrease to zero).

The outcome of this example gives an intuitive impression of what about to expect if we turn from our special function to an arbitrary one, as we shall do next. ■

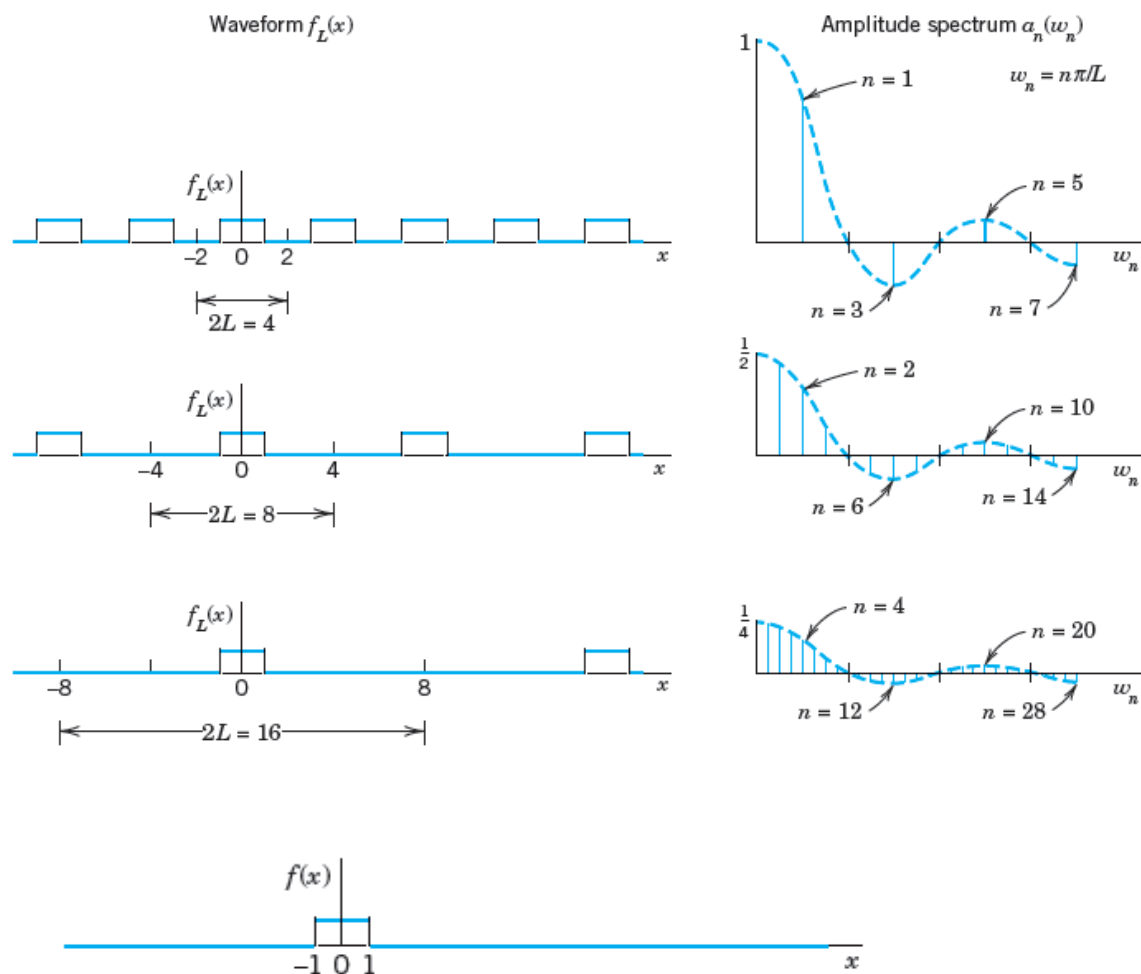


Fig. 280. Waveforms and amplitude spectra in Example 1

From Fourier Series to Fourier Integral

We now consider any periodic function $f_L(x)$ of period $2L$ that can be represented by a Fourier series

$$f_L(x) = a_0 + \sum_{n=1}^{\infty} (a_n \cos w_n x + b_n \sin w_n x), \quad w_n = \frac{n\pi}{L}$$

and find out what happens if we let $L \rightarrow \infty$. Together with Example 1 the present calculation will suggest that we should expect an integral (instead of a series) involving $\cos wx$ and $\sin wx$ with w no longer restricted to integer multiples $w = w_n = n\pi/L$ of π/L but taking *all* values. We shall also see what form such an integral might have.

If we insert a_n and b_n from the Euler formulas (6), Sec. 11.2, and denote the variable of integration by v , the Fourier series of $f_L(x)$ becomes

$$f_L(x) = \frac{1}{2L} \int_{-L}^L f_L(v) dv + \frac{1}{L} \sum_{n=1}^{\infty} \left[\cos w_n x \int_{-L}^L f_L(v) \cos w_n v dv + \sin w_n x \int_{-L}^L f_L(v) \sin w_n v dv \right].$$

We now set

$$\Delta w = w_{n+1} - w_n = \frac{(n+1)\pi}{L} - \frac{n\pi}{L} = \frac{\pi}{L}.$$

Then $1/L = \Delta w/\pi$, and we may write the Fourier series in the form

$$(1) \quad f_L(x) = \frac{1}{2L} \int_{-L}^L f_L(v) dv + \frac{1}{\pi} \sum_{n=1}^{\infty} \left[(\cos w_n x) \Delta w \int_{-L}^L f_L(v) \cos w_n v dv + (\sin w_n x) \Delta w \int_{-L}^L f_L(v) \sin w_n v dv \right].$$

This representation is valid for any fixed L , arbitrarily large, but finite.

We now let $L \rightarrow \infty$ and assume that the resulting nonperiodic function

$$f(x) = \lim_{L \rightarrow \infty} f_L(x)$$

is **absolutely integrable** on the x -axis; that is, the following (finite!) limits exist:

$$(2) \quad \lim_{a \rightarrow -\infty} \int_a^0 |f(x)| dx + \lim_{b \rightarrow \infty} \int_0^b |f(x)| dx \quad \left(\text{written } \int_{-\infty}^{\infty} |f(x)| dx \right).$$

Then $1/L \rightarrow 0$, and the value of the first term on the right side of (1) approaches zero. Also $\Delta w = \pi/L \rightarrow 0$ and it seems *plausible* that the infinite series in (1) becomes an

integral from 0 to ∞ , which represents $f(x)$, namely,

$$(3) \quad f(x) = \frac{1}{\pi} \int_0^{\infty} \left[\cos wx \int_{-\infty}^{\infty} f(v) \cos wv \, dv + \sin wx \int_{-\infty}^{\infty} f(v) \sin wv \, dv \right] dw.$$

If we introduce the notations

$$(4) \quad A(w) = \frac{1}{\pi} \int_{-\infty}^{\infty} f(v) \cos wv \, dv, \quad B(w) = \frac{1}{\pi} \int_{-\infty}^{\infty} f(v) \sin wv \, dv$$

we can write this in the form

$$(5) \quad f(x) = \int_0^{\infty} [A(w) \cos wx + B(w) \sin wx] \, dw.$$

This is called a representation of $f(x)$ by a **Fourier integral**.

It is clear that our naive approach merely *suggests* the representation (5), but by no means establishes it; in fact, the limit of the series in (1) as Δw approaches zero is not the definition of the integral (3). Sufficient conditions for the validity of (5) are as follows.

THEOREM 1

Fourier Integral

If $f(x)$ is piecewise continuous (see Sec. 6.1) in every finite interval and has a right-hand derivative and a left-hand derivative at every point (see Sec. 11.1) and if the integral (2) exists, then $f(x)$ can be represented by a Fourier integral (5) with A and B given by (4). At a point where $f(x)$ is discontinuous the value of the Fourier integral equals the average of the left- and right-hand limits of $f(x)$ at that point (see Sec. 11.1). (Proof in Ref. [C12]; see App. 1.)

Applications of Fourier Integrals

The main application of Fourier integrals is in solving ODEs and PDEs, as we shall see for PDEs in Sec. 12.6. However, we can also use Fourier integrals in integration and in discussing functions defined by integrals, as the next example.

EXAMPLE 2

Single Pulse, Sine Integral. Dirichlet's Discontinuous Factor. Gibbs Phenomenon

Find the Fourier integral representation of the function

$$f(x) = \begin{cases} 1 & \text{if } |x| < 1 \\ 0 & \text{if } |x| > 1 \end{cases} \quad (\text{Fig. 281})$$

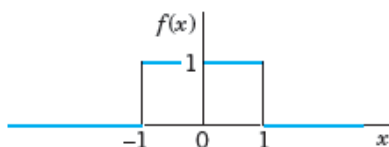


Fig. 281. Example 2

Solution. From (4) we obtain

$$A(w) = \frac{1}{\pi} \int_{-\infty}^{\infty} f(v) \cos wv \, dv = \frac{1}{\pi} \int_{-1}^1 \cos wv \, dv = \frac{\sin wv}{\pi w} \Big|_{-1}^1 = \frac{2 \sin w}{\pi w}$$

$$B(w) = \frac{1}{\pi} \int_{-1}^1 \sin wv \, dv = 0$$

and (5) gives the *answer*

$$(6) \quad f(x) = \frac{2}{\pi} \int_0^{\infty} \frac{\cos wx \sin w}{w} dw.$$

The average of the left- and right-hand limits of $f(x)$ at $x = 1$ is equal to $(1 + 0)/2$, that is, $\frac{1}{2}$.

Furthermore, from (6) and Theorem 1 we obtain (multiply by $\pi/2$)

$$(7) \quad \int_0^{\infty} \frac{\cos wx \sin w}{w} dw = \begin{cases} \pi/2 & \text{if } 0 \leq x < 1, \\ \pi/4 & \text{if } x = 1, \\ 0 & \text{if } x > 1. \end{cases}$$

We mention that this integral is called **Dirichlet's discontinuous factor**. (For P. L. Dirichlet see Sec. 10.8.)

The case $x = 0$ is of particular interest. If $x = 0$, then (7) gives

$$(8^*) \quad \int_0^{\infty} \frac{\sin w}{w} dw = \frac{\pi}{2}.$$

We see that this integral is the limit of the so-called **sine integral**

$$(8) \quad \text{Si}(u) = \int_0^u \frac{\sin w}{w} dw$$

as $u \rightarrow \infty$. The graphs of $\text{Si}(u)$ and of the integrand are shown in Fig. 282.

In the case of a Fourier series the graphs of the partial sums are approximation curves of the curve of the periodic function represented by the series. Similarly, in the case of the Fourier integral (5), approximations are obtained by replacing ∞ by numbers a . Hence the integral

$$(9) \quad \frac{2}{\pi} \int_0^a \frac{\cos wx \sin w}{w} dw$$

approximates the right side in (6) and therefore $f(x)$.

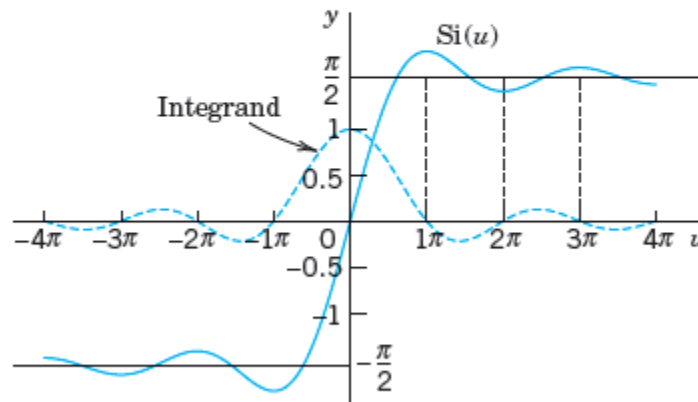


Fig. 282. Sine integral $\text{Si}(u)$ and integrand

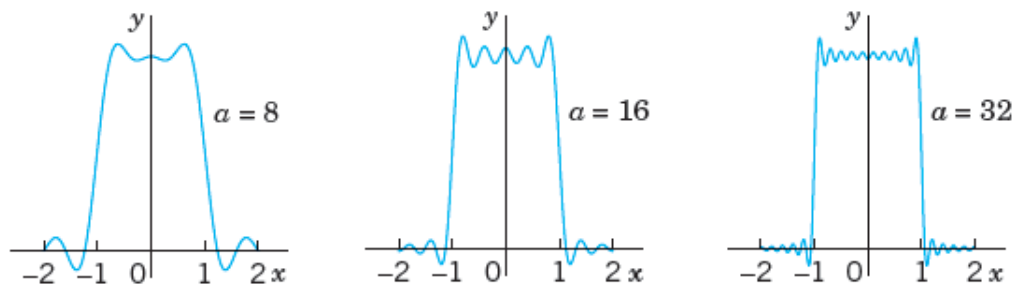


Fig. 283. The integral (9) for $a = 8, 16$, and 32 , illustrating the development of the Gibbs phenomenon

Figure 283 shows oscillations near the points of discontinuity of $f(x)$. We might expect that these oscillations disappear as a approaches infinity. But this is not true; with increasing a , they are shifted closer to the points $x = \pm 1$. This unexpected behavior, which also occurs in connection with Fourier series (see Sec. 11.2), is known as the **Gibbs phenomenon**. We can explain it by representing (9) in terms of sine integrals as follows. Using (11) in App. A3.1, we have

$$\frac{2}{\pi} \int_0^a \frac{\cos wx \sin w}{w} dw = \frac{1}{\pi} \int_0^a \frac{\sin(w + wx)}{w} dw + \frac{1}{\pi} \int_0^a \frac{\sin(w - wx)}{w} dw.$$

In the first integral on the right we set $w + wx = t$. Then $dw/w = dt/t$, and $0 \leq w \leq a$ corresponds to $0 \leq t \leq (x + 1)a$. In the last integral we set $w - wx = -t$. Then $dw/w = dt/t$, and $0 \leq w \leq a$ corresponds to $0 \leq t \leq (x - 1)a$. Since $\sin(-t) = -\sin t$, we thus obtain

$$\frac{2}{\pi} \int_0^a \frac{\cos wx \sin w}{w} dw = \frac{1}{\pi} \int_0^{(x+1)a} \frac{\sin t}{t} dt - \frac{1}{\pi} \int_0^{(x-1)a} \frac{\sin t}{t} dt.$$

From this and (8) we see that our integral (9) equals

$$\frac{1}{\pi} \text{Si}(a[x + 1]) - \frac{1}{\pi} \text{Si}(a[x - 1])$$

and the oscillations in Fig. 283 result from those in Fig. 282. The increase of a amounts to a transformation of the scale on the axis and causes the shift of the oscillations (the waves) toward the points of discontinuity -1 and 1 . ■

